

PARTICULAR SPECIFICATION P6A

APPENDIX P6A – Digestion Pre-treatment and Sidestream Nitrogen Removal Process Control Narrative

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P6A.1 INTRODUCTION

P6A.1.1 Process Overview

The Digestion Pre-treatment facility and Sidestream Nitrogen (N) Removal facility at TWRP are to operate in a fully automatic and autonomous manner with the minimum of human intervention required. The control system shall facilitate this, along with ensuring the facility operates and shuts down in an inherently safe manner.

This PCN shall be read in conjunction with the Standard Control Features PCN which details general requirements that are common to the entire TWRP site

In addition to the main thermal hydrolysis process, the scope of process control for Digestion Pre-treatment plant shall cover the associated systems (in this Contract) including THP feed pumps, Hydrolysed sludge cooling, Hydrolysed sludge transfer/recirculation line to digester, Low Pressure Hot Water (LPHW) supply, Steam desuperheater and Instrument air package. The process control narratives and the functional design specifications for the THP system and associated systems shall be developed by the THP specialist supplier and be incorporated into the overall plant control system.

The term THP system shall be used interchangeably with digestion pre-treatment facility throughout this document, this includes the main THP process and the supporting systems which are part of the Digestion Pre-treatment Contract scope.

P6A.1.2 Control System Software Programming Roles

Refer to Particular Specification P6 ICA Works for the definitive allocation of software programming scope.

In general terms, parties responsible for software programming are indicated in Table 1-1. Classification of PLC's is also given in this table. Based on early vendor evaluation, the THP system typically comes with package unit PLC. Considering the Contract's scope and the interfacing requirements, it is proposed that one Master THP PLC (hot standby) shall be provided to interface with the TWRP's other plant PLCs and also provide the high-level control for individual THP unit PLC and the supporting equipment in the Contract's scope (e.g., instrument air system). For Sidestream N Removal system, one PLC(hot standby) shall be provided to interface with the TWRP's other plant PLCs and control of overall Sidestream N removal facility including the deammonification process of the four deammonification reactors. Another hot-standby PLCs shall be provided as a master control for Sidestream N Removal blowers.

Table 1-1: Software Programming Allocation

Process / Plant Area	Control Philosophy and Software FDS (Functional Design Specification)	PLC classification (as per G6.17)	PLC Programming, Testing & Commissioning	Programming, Testing & Commissioning for On-Site HMI's	Plant SCADA Programming, Testing & Commissioning
Individual THP Package Units	THP System Supplier (C4B)	Vendor Package Design and Built PLC, supplied as part of Vendor package	THP System Supplier	THP System Supplier	S.O. (with input from C4B)
Master Control of THP system (including THP feed pumps, Hydrolysed Sludge coolers, digesters feed pumps etc.)	C4B, THP system supplier (with input from S.O. and C4A)	Vendor Package Design and Built PLC, supplied as part of Vendor package	C4B (with interface input from S.O. and C4A)	THP System Supplier	S.O. (with input from C4B)
Master Control of Sidestream N Removal	Sidestream N-Removal System Supplier	Vendor Package Design and Built PLC, supplied as part of Vendor package	Sidestream N System Supplier	Sidestream N System Supplier	S.O. (with input from C4B)
Master Control of Sidestream N Aeration Blower	Sidestream N-Removal Aeration Blower manufacturer	Machine PLC	Sidestream N-Removal Aeration Blower manufacturer	Sidestream N-Removal Aeration Blower manufacturer	S.O. (with input from C4B)
Unit Control of Sidestream N Aeration Blower	Sidestream N-Removal Aeration Blower manufacturer	Machine PLC	Sidestream N-Removal Aeration Blower manufacturer	Sidestream N-Removal Aeration Blower manufacturer	S.O. (with input from C4B)

P6A.1.3 Digester Pre-treatment Interface with C4A

There are multiple interfaces between Biosolids & Digesters contractor (C4A contractor) and Digester Pre-treatment contractor (Contractor) to enable the THP PLC to control the THP and associated systems. Some of the signals are referenced in the relevant sections of this document.

The interface signals required included here are a minimum requirement, the full list of interfaces required will be developed with the specialist supplier of the THP plant (as well as C4A's pre-dewatering system and Silo specialist):

From C4A to C4B:

- DWAS silo level
- DWAS silo weight
- DWAS silo discharge slide gate status
- DWAS silo status
- DWAS sludge concentration (% dry solid content)
- Overall Digesters' Bulk temperatures
- Feeding Digester's mixing temperature
- Feeding Digester's HS flow rates
- HS digester feed route available
- FA digester discharge route available
- LPHW Tank level
- LPHW Tank Temperature
- GWRF LPHW route available/ready to receive signal
- HWR to GWRF target temperature set point
- HWR to GWRF temperature
- HWR to GWRF target flow set point
- HWR to GWRF Pressure
- Plant water G2D status
- TCD wet well status

From C4B to C4A:

- DWAS silo discharge slide gate open / close request
- HS digester feed flowrate and setpoint
- THP status
- HWR to GWRF flow rate
- HWR to GWRF temperature
- IA status
- IA receiver pressure
- Process Gas outlet temperature
- HS temperature
- HS digester feed pump status
- THP unit status
- THP Available Capacity

P6A.1.4 Interface with IWMF

There are multiple ICA interfaces related to steam supply with the IWMF to monitor and control. These interfaces signals will be exchanged with IWMF through C4B PLC and Plant PLC. The RIO panel (by Other contractors) to interface with IWMF will be located at the end of the IWMF DWS Transfer Conveyors, within the TWRP secure boundary, power supply will be by TWRP.

These signals are detailed in:

Particular Specifications P6.6.8 and in P6A.3.6

Summaries of the signals are contained in the relevant sections of this document.

P6A.1.5 Sidestream N-Removal Facility Interfaces with C4A, C3B1 and C3B2

The Sidestream N-Removal Facility interfaces with C4A for the input fluid streams and to C3B1 and C3B2 for the output streams.

The intention is that Sidestream N-Removal Facility normally operates independently from a process control perspective, and only monitoring and emergency shutdown signals are required with the other process units.

P6A.2 THP OVERVIEW

Figure 2-1 shows the overall biosolids plant schematic up to the digesters. The main objective of the THP system is to hydrolyse up to 50% of the liquid modules' solids during design capacity.

The remaining primary sludge from both ILM and DLM are thickened before being fed to the digesters. The digesters also receive greasy waste and food waste. The digesters will operate in mesophilic conditions, therefore the hydrolysed sludge from THP shall be cooled before routing to the digesters.

The Digestion Pre-treatment facility is a Thermal Hydrolysis process (THP), typically a semi-batch process. The THP system receives the pre-dewatered sludge cake from the silo and pre-heats the feed, this is then followed by reactors which provide the conditions required for the thermal hydrolysis reactions, including high temperature by introducing direct steam contact. The rheology and biodegradability of the resulting sludge stream shall be improved. The high temperature product sludge shall be cooled via heat exchangers before sending to the digester feed line. The scope of the THP system starts from the THP feed cake pumps. These pumps shall interface with the DWAS Silos controls to provide the required feed flowrate to the main THP process units. The THP system controls up to the HS digester feed pumps and additional recirculation pumps which interfaces with the Digesters' feed line and step-feeding cycle sequences. This also includes the Digester temperature control which shall be cascaded with the HS cooling system as described in Sections P6A.4. In addition to pre-treatment of sludge, the THP also functions to control the heat recovery from HS cooling and provide hot water to the Greasy Waste facilities.

The following section P6A.3 describes the THP system interfacing requirements and controls expected to be developed by the THP provider. Refer to section P6A.5 for details on the digester feeding sequence required. The THP system's boundary is conceptually depicted in Figure 2-2, showing the upstream, downstream system as well as various utilities interfaces. The control philosophy of the THP system shall be carefully developed with consideration of the interfaces and must be fully integrated to achieve stable, efficient, and safe operation.

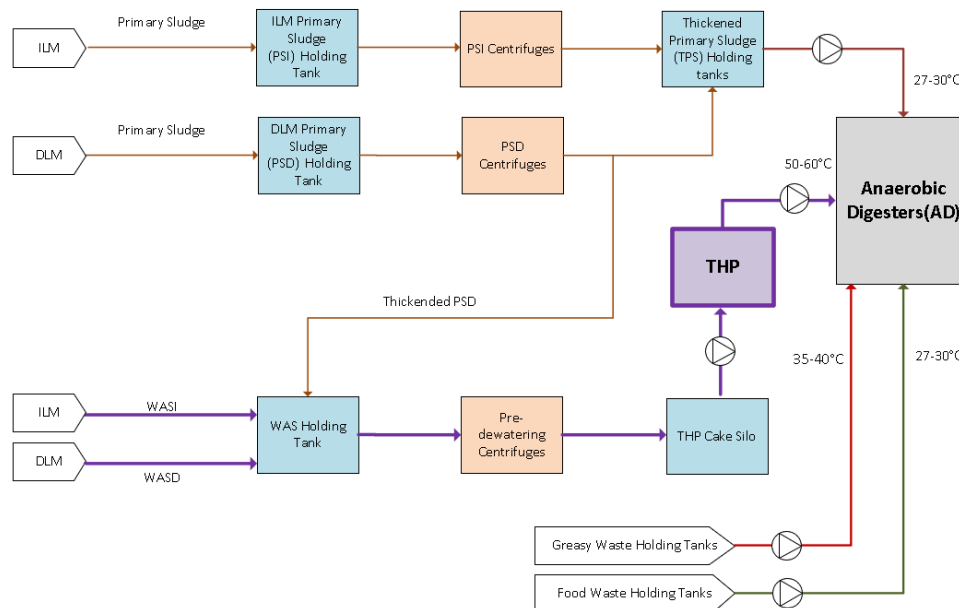


Figure 2-1 Overall Block Diagram of Digester Biosolid feed system

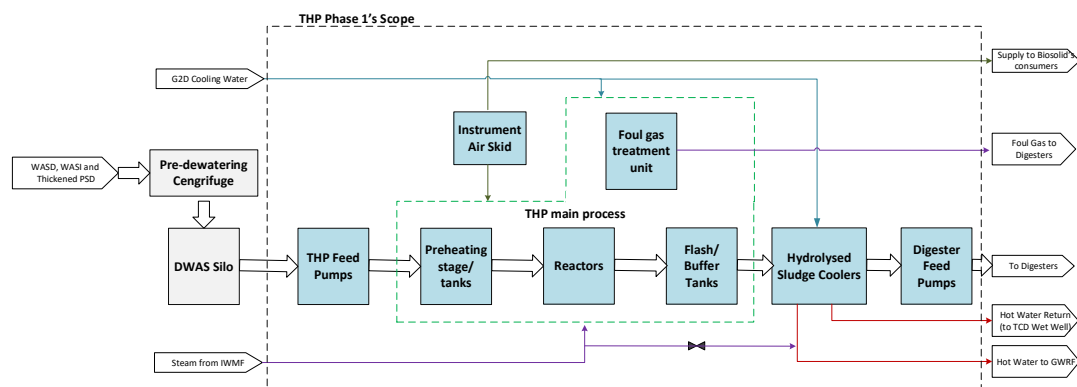


Figure 2-2 Process schematics of THP system

P6A.3 THERMAL HYDROLYSIS PROCESS

P6A.3.1 Overview

The operation of the Thermal Hydrolysis System will be determined by the THP specialist supplier. The operation and control of the systems must be fully integrated to achieve stable, efficient, and safe operation with minimal manual intervention.

The process control narratives and the functional design specifications for the THP system shall be developed by the THP specialist supplier and be incorporated into the overall plant control system. The PCN shall be developed to a level of detail suitable for programming and configuring the process control system. The PCN shall be concise and includes logic diagrams, flow charts or sequence list to be read in conjunction with the control narratives.

The Standard details Appendix P6B shall be the basis for developing the interfaces between LCS/PLC/SCADA for motor drives, instruments, and controller control loops. Equipment within package shall follow the same philosophy for consistency to maximum possible extend. In this section the control of the interfaces and the expected high-level controls of the THP system will be described.

The PCN is described the operation and control philosophy of the THP including the future 4th THP unit and related feed pumps and heat exchangers. The future 4th THP is not included within the scope of this Contract, hence the future THP unit and the related equipment shall be set as OUT OF SERVICE on SCADA.

P6A.3.2 Control Objectives

TWRP THP System shall be fully automated with minimum operator intervention. The control system design shall achieve the following:

- Achieve consistent HS feed to the digester system;
- Maintain safe, stable and efficient operation to achieve the degree of hydrolysis required;
- Provide flexibility in control system to enable operator to select the number of THP units operating and shall have various operating modes to allow for turndown capacity and performance/energy optimisation;
- Reduce manpower through higher-level plant automation.

P6A.3.3 Operator Settings

The THP PCN shall define the list of operator adjustable settings required for a fully functional Thermal Hydrolysis system. All essential monitoring parameters shall be shown in HMI and SCADA simultaneously and acknowledgement of alarms/trips shall be made available in HMI and SCADA. All relevant information including level of alarms and descriptions shall be provided to the S.O. for SCADA development.

The authority level with permission to change settings shall be proposed for each setting.

- With four(4) THP trains, it is envisaged that the THP units shall be able to operate in either:
 - N-1 scenario (Default): This shall be the default setting during design flowrate i.e., 4 units normally in operation, and 3 units to receive the full flow when one unit is offline.
 - N+1 scenario: three (3) DUTY trains and one(1) STANDBY train; or
 - During initial operations or turndown conditions, it shall be able to operate in two (2) Duty, two(2) standby or One (1) duty, three(3) standbys.
- Depending on the operating scenarios, the units will be assigned as DUTY1, DUTY2/STANDBY1, DUTY3/STANDBY2, DUTY4/STANDBY3.

The duty assignments of each THP Unit will be manually selected from the SCADA.

As a minimum, the following operator settings shall be adjustable and set on SCADA:

- Minimum and Maximum flow capacity of individual THP Units
- The operating level settings of the DWAS silos
- Individual THP HS feed flow setpoint
- If there several modes of operations, the selections shall be made available on SCADA
- Primary timer setpoints for individual THP sequences

And the following shall be initiated from SCADA

- Initiating Start-up of individual THP units
- Initiating Stop sequence of individual THP units (or/and PAUSE if applicable)
- Putting individual THP units OUT OF SERVICE

P6A.3.4 Normal Operation

The Contractor and its appointed specialist shall propose a list of THP live status indicators to be displayed on SCADA, this shall provide Operator an overview of the performance and efficiency of the system. All units proposed shall be submitted and approved. As a minimum the following shall be included:

- List of status of Individual THP units: The status of individual THP units shall be shown on SCADA (e.g., IN SERVICE, OUT OF SERVICE). The status of individual reactors shall be displayed on SCADA e.g., FILLING, DISCHARGING. The total runtime of each THP units shall also be displayed on SCADA.
- Steam consumption per tDs
- Cooling water consumption
- Power consumption per tDs. Power consumption shall be provided as input to TWRP's Power Monitoring System (PMS) by C5A contractor.
- Instrument Air consumption

The Contractor's control specialist shall work closely with the TWRP team for integrating the THP and SCADA interfaces.

P6A.3.5 THP Feed Pump Control

N.B. The following control description is indicative of the level of automation required. The Contractor shall provide a control philosophy from the specialist supplier(s) and shall program the controls and monitoring required for their system in the Contractor's PLCs and HMIs.

There will be two (2) THP feed cake pumps dedicated to each THP train. The THP feed pumps for the DUTY THP train shall be assigned as DUTY and STANDBY respectively. The duty assignments of each THP feed pump will be automatically assigned in line with the overall duty rotation strategy or can be manually selected from the SCADA system.

When in operation the THP process vessel shall accept flows from the THP Feed Pumps. The THP feed pump shall be provided to deliver the THP cake or pre-dewatered sludge from two (2) DWAS Silos to the THP units. Silo 1 shall be dedicated to 1 THP feed pump from each THP train, while Silo 2 shall serve the other THP feed pump from each THP train. This minimises the number of elbow and bends in process piping while providing the required redundancy.

It is envisaged that these pumps will be VFD type and operates based on pressure or flow control mode. The pump controllers shall react to the DWAS silo levels as well as additional trim from the THP levels. Confirmation of an open flow path in its respective THP units and HS digesters feed route shall be required before the feed pumps are started. It is envisaged that the HS feed pump should be initiated as part of the THP unit start-up sequence. The pumps shall receive permissive signals from level in the DWAS Silos and permissive to start signal from the THP. Section P6A.5.4b) explains the typical feed flow attenuation control based on Silo level.

As the feed flow into THP units could be less than the feed flow output from each THP unit (due to the addition of steam), the THP Feed pump flow setpoint may be trimmed from the output i.e., HS feed flow setpoint. The trimmed factor could be cascaded based on the feedback from THP vessel level deviation. The control details shall be developed by THP specialist, to ensure the attenuation between silo, THP units and digester feeding. Additionally, if THP system requires precise control of THP feed %dry solid content, optimized plant water dilution may be added as a trim for the feed cake pump control. When flushing is critical as part of start-up /shutdown operations, automatic flushing shall be implemented.

P6A.3.6 Steam Supply

The steam is supplied from the IWMF for the operation of the THP. There are one (1) DUTY and one(1) STANDBY incoming steam pipelines from IWMF battery limit with intention to operate in N-1 philosophy. Both pipelines shall be online under normal operating conditions, except during turndown flowrates or when one pipeline is under maintenance. Prior to entering the THP units, a desuperheater shall function to bring the steam as close to saturation point as possible.

Typically, the steam desuperheater shall consist of a Temperature control valve at the water supply line which shall inject pressurised cooling water into the live steam. The Contractor shall develop the control philosophy and control and monitoring program for Steam flow control into THP units, as well as steam conditioning to the required saturation temperature/pressure. It is envisaged that the steam desuperheater system will require a booster pump for the water injection as the supply pressure from G2D ring main is much lower than the steam pressure.

The steam control system shall be developed for automatic control to required turndown flowrate. Additionally, safeguarding of high-pressure interlocks shall be implemented to protect THP units. Any emergency shutdown or sudden reduction in capacity shall be communicated to IWMF.

In case of unavailability of heating energy from HS stage 1 cooling heat exchangers for generating LPHW supply, some amount of Steam shall be used to provide the heating via Steam Sparger (inline jet heating). This is further explained in P6A.3.10.

The steam consumption in terms of tonne/ds shall be recorded and displayed on SCADA.

To facilitate the operation the IWMF will provide the following signals:

- IWMF Steam Available
- IWMF Steam Pressure
- IWMF Steam Temperature
- IWMF Steam Mass Flow

TWRP will provide the following signals:

- TWRP Steam Route Available
- TWRP Steam Controlled Stop Request
- TWRP Steam Emergency Shutdown

P6A.3.7 HS Cooling Heat Exchangers

The Hydrolysed Sludge from THPs are envisaged to be at high temperature, thus cooling is required prior to feeding to the Digesters.

The heat shall be recovered from first stage cooler for hot water usage at Greasy Waste Facilities. The second stage cooler shall provide fine-tuning to the required HS temperature setpoint which shall be trimmed as required to achieve required digesters' bulk temperature setpoint. The flow of hydrolysed sludge from the THP units shall be passed through two-stage cooling before being fed to the digesters.

As a summary, the following control objective shall be met:

- First Stage Heat Exchanger (including steam spargers and bypass flow controls);
 - Provide hot water for the GWRF – pipe jacketing and LPHW spray/wash water;
 - Minimise the use of cooling water
 - Provide sufficient flow at 50°C for the GWRF (via common bypass control valve)
 - Minimise supplemental steam consumption
- Second Stage Heat Exchanger
 - Maintain the digester temperature set points;
 - Minimise the use of cooling water

Each THP unit will discharge HS into a DUTY / DUTY two stage cooling system. The Standby cooling will be provided at the standby THP unit. The HS Cooling Heat Exchangers (HEX) are assigned as either STAGE 1: DUTY, STAGE 2: DUTY or STAGE 1: STANDBY, STAGE 2: STANDBY.

The duty assignments of each HEX will not be automatically assigned in line with the overall duty rotation strategy, the duty assignments shall be dependent on the assignment of the upstream THP unit or can be manually selected from the SCADA system.

The operator shall set:

- The individual HEX status i.e., IN SERVICE or OUT OF SERVICE;
- The GW hot water target temperature setpoint;
- The GW hot water (LPHW) target flowrate setpoint;
- The initial HS cooler outlet temperature setpoint.

The overall HEX control is shown schematically in Figure 3-1 1 depicting the expected control requirement, it is the Contractor's responsibility to further develop the schematic to be in-line with the selected design and equipment. The following sections describe the expected controls in further details.

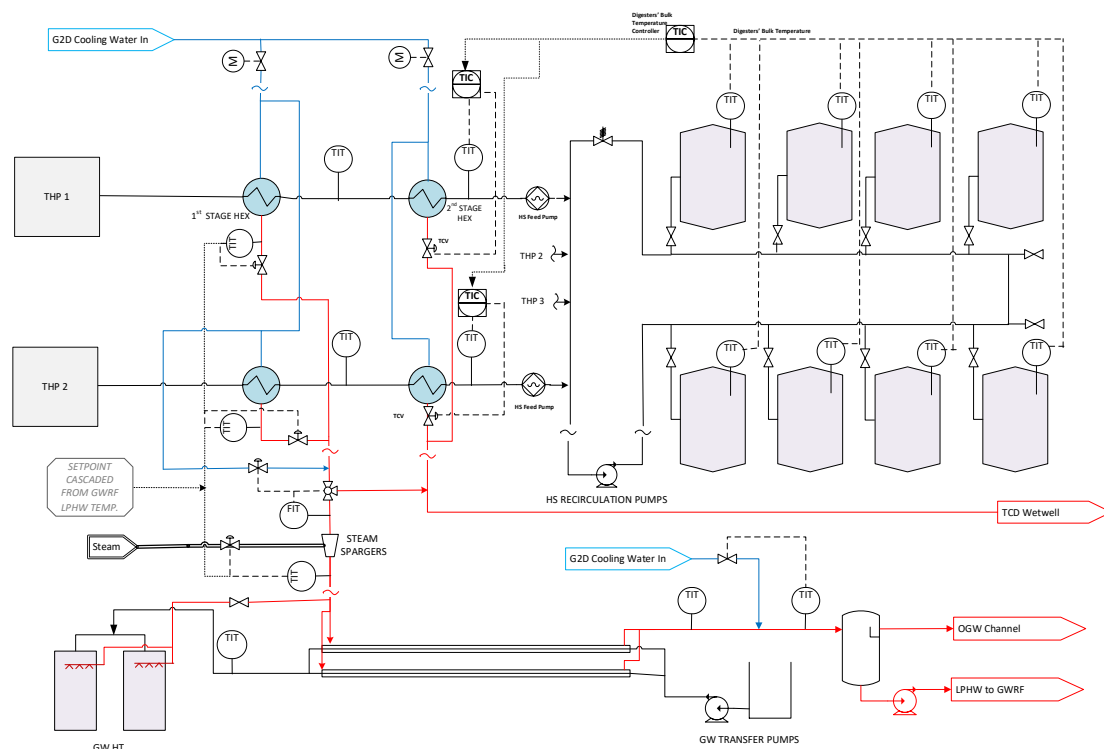


Figure 3-1 Preliminary control schematic of Hydrolysed Sludge Cooling and Heat Recovery for Hot Water supply to Greasy Waste

P6A.3.8 First stage HS Cooling Heat Exchanger

The first stage of heat exchanger will be controlled to achieve a set temperature of hot water to the GWRF, while also cooling the HS to an intermediate temperature.

It is likely that recirculation of hydrolysed sludge will be required for achieving turbulent flow regime in the tube side to promote high heat transfer efficiencies. The control philosophy shall be further developed by the Contractor with input from the THP/heat exchanger specialist.

The cooling water Temperature Control Valve (TCV) at the outlet of each Stage 1 HEX shall modulate to achieve the GWRF HW Temperature setpoint. The initial setpoint of the GWRF LPHW Temperature shall be set manually from the SCADA system. When setpoint is in auto mode the setpoint shall be trimmed based on the temperature feedback of the LPHW arriving at the GWRF. The dead band of the trim range shall be optimised to ensure the cooling of hydrolysed sludge in the first stage HEX is not compromised. The typical arrangement and expected controls of the first stage HEX and the common bypass flow and supplemental heating controls are depicted in Figure 3-11. The following sections P6A.3.10 and P6A.3.11 describes the associated supplemental heating system and the bypass and flow control respectively.

P6A.3.9 Second stage HS Cooling Heat Exchanger

The cooling water flowrate for the second stage heat exchanger will be controlled to achieve the HS cooler outlet temperature setpoint. The HS cooler outlet temperature setpoint ($T_{HS_setpoint}$) shall be an output from the Digester Bulk temperature controller, further described in Section P6A.4. The digesters' bulk temperature controller shall be part of the THP's scope.

The TCV on the cooling water stream of each Second stage HEX shall modulate to achieve the HS Cooler outlet Temperature setpoint. The inlet to the second stage HEX shall be the intermediate temperature from the outlet of the first stage HEX. Since the intermediate HS temperature is not controlled, variation is expected because of any imbalance in the THP buffer/flash tank outlet's temperature. Therefore, the temperature control is important to maintain the outlet temperature by adjusting the cooling water flowrate. The TCV and temperature controller shall be fast to react to any changes to the $T_{HS_setpoint}$ during the feeding cycle. The high limit settings to $T_{HS_setpoint}$ is also explained in P6A.4.3. In initial years, and depending on the digester temperature setpoint, the $T_{HS_setpoint}$ may be close to the intermediate HS temperature (and exceeds turndown control of second stage HEX) e.g., 60-70°C, there shall be option to bypass the second heat exchanger and the digester temperature controller will be set to Manual. Care shall be taken in the intermediate HS temperature T_{HS_interm} to ensure the high temperature alarm and trip settings (could be an in-built thermal sensor) shall protect the design temperature of the HS feed pumps' stator. Further description on high temperature monitoring at the digester feed nozzle is explained in P6A.4.3.

The Contractor shall develop the control narratives for the heat exchangers temperature controller and cooler's recirculation line on the sludge side. The control philosophy shall be developed with input from heat exchanger specialist. If provided, the sludge cooler pumps and recirculation line controls shall provide adequate controls to ensure the hydrolysed sludge digester feeding system can operate smoothly.

The temperature control is shown schematically in Figure 3-11 depicting the expected control requirement.

P6A.3.10 Steam sparger

Steam sparger or (inline steam injection) shall be provided for supplemental heating of hot water to GW facilities. In the occasion where the heat exchangers fail to achieve the required heating water temperature or flowrate for supply to GW facilities, the Steam Sparger system (consisting of TCVs and inline steam nozzles) shall be initiated.

There is a dedicated temperature transmitter on the common hot water supply line downstream of the steam sparger (as shown in Figure 3-2) . This temperature transmitter together with the steam sparger's temperature control valve (TCV) shall form the temperature controller for the hot water supplemental heating.

The steam sparger shall be started only there is confirmed low temperature alarm (e.g., lower than 45°C) registered for more than a certain pre-set time delay. Option shall be provided to prompt Operator before the steam sparger is initiated. The steam sparger's functional description and controls shall be further developed by the Contractor with input from steam sparger supplier. The TCV and temperature controller shall be efficient in modulating the steam injection rate in order to achieve steady hot water supply temperature while minimising steam consumption. When high temperature is detected, steam sparger controller shall automatically ramp down the steam flow, and if temperature continue to rise till HH setpoint, the steam sparger shall automatically switch off.

Due to the high pressure of steam, a pressure reducing valve is expected upstream of the TCV. A safe and efficient control shall be developed for starting up, ramping of the control valves. Adequate PID tuning controls shall be provision for ensuring smooth commissioning and operation.

The Contractor shall develop proper safety interlocks and safeguarding controls while considering the steam and Hot water pipeline design pressure and temperature. The steam sparger shall be interlocked to stop when low or no flow alarm in the hot water stream is detected.

P6A.3.11 First Stage Heat Exchanger Common Bypass Line Flow Control

The combined hot water flowrate from all the Duty 1st stage Heat Exchangers is measured by a Flow Meter at the common outlet (as in Figure 3-2) .

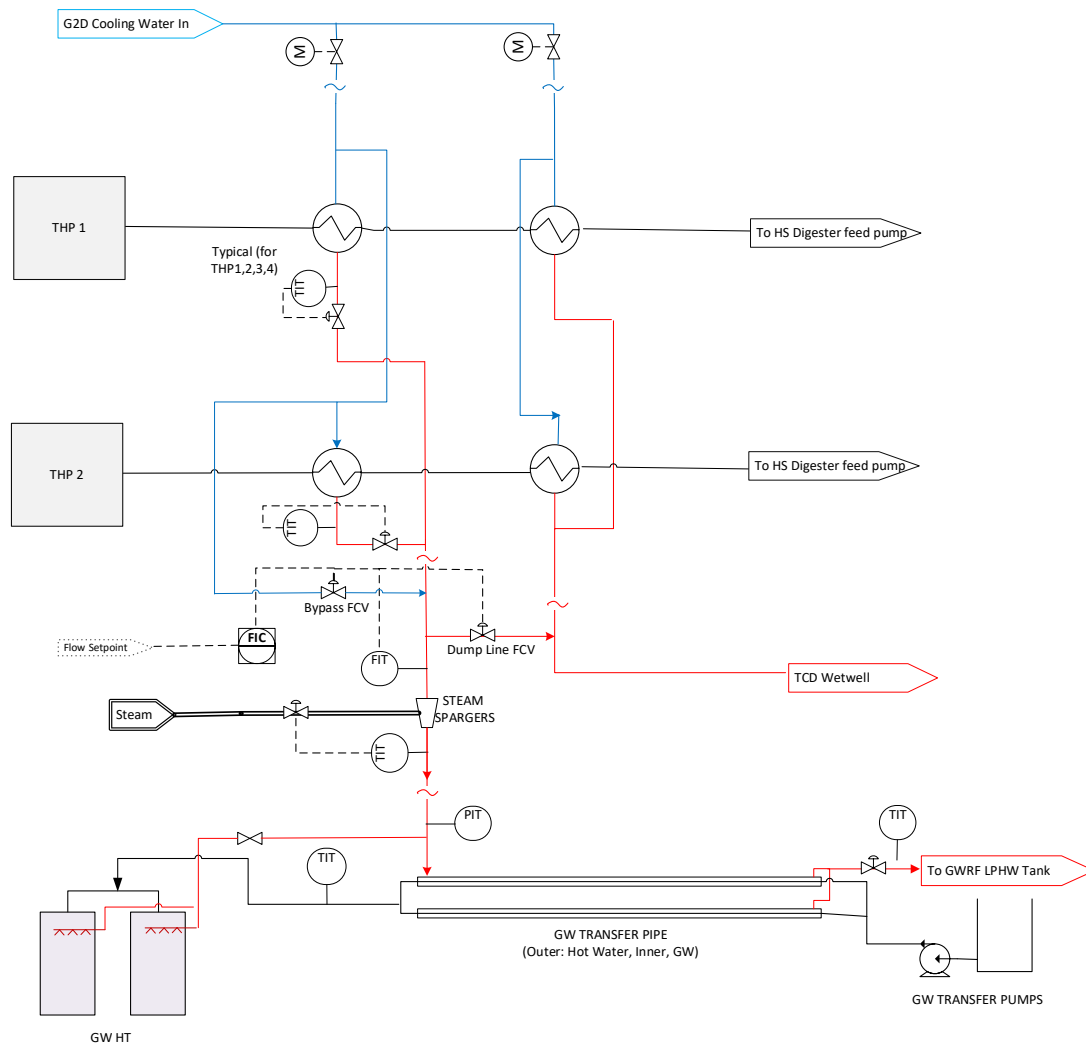


Figure 3-2 Schematic of First Stage Heat exchanger outlet Bypass and Dump line flow controller

Under normal operation, it is expected that a percentage of the flow shall be dumped via the dump valve (e.g., 5-10%). This serves as a buffer for the heat exchanger's temperature control to counteract any imbalance load from the cooling of hydrolysed sludge. The flow controller shall modulate the dump valve towards the close position when the flow detected is lower than the setpoint. In the occasion where the heat exchangers fail to achieve the required heating water flowrate for GW's LPHW, and the dump valve is already fully closed, the flow controller will detect the deviation and initiates the bypass line FCV to activate.

It is expected that the dump valve and heat exchanger are designed to cater for the imbalance load from THP, during design flowrates. The bypass FCV shall only operate when one or more THP units are unavailable. Either the dump valve or the bypass valve shall be open at any one time.

The bypass line flow control shall be designed to handle a high turndown ratio to meet various scenarios. It is envisaged that two(2) different control valve sizes FCV1 and FCV 2 will be required for the turndown flow control. The flow controller shall be based on split ratio control in this case. The following illustrates the possible scenarios and CV openings of the bypass and dump line FCV.

Table 3-1: Possible Scenarios and CV Openings of Bypass and Dump Line FCV

Scenarios	GW HW Flow setpoint m ³ /h	Bypass FCV (CV %)	Dump line FCV (CV %)
1) GW Operating hours - Design THP capacity	105	FCV1: 0% FCV2: 0%	0-10 % (0-10m ³ /h)
2) GW Operating hours THP 50% turndown capacity	105	FCV1: 0% FCV2: 50% (e.g., 50m ³ /h)	0%
3) GW Operating hours THP 85% turndown capacity	105	FCV1: 70% (e.g., 16m ³ /h) FCV2: 0%	0%

It is expected that during turndown capacity (or when with HEX bypass), the comingled stream temperature will drop below setpoint and this shall initiate the steam sparger. The Operator shall be able prompted to allow the initiation of steam sparger.

The GWRF operates daily from 8:00 to 6:00pm (Mon to Sat). During non-operating hours, the GWRF shall stop receiving hot water and the GW transfer line will be offline as well. Only intermittent hot water spray is required at the GW holding tanks near to Digesters. It is expected to be few minutes once every few hours. During non-operating hours, the GW hot water flow setpoint shall be 0 m³/h. The dump line FCV shall divert all the hot water from first stage heat exchanger to the TCD wet well. Note that the dump line FCV shall be able to maintain a backpressure of 3-4barg during the full flow diversion. This shall allow the intermittent hot water spray to function at the required pressure. The following illustrates the possible CV openings of the bypass and dump line FCV during non-operating hours of GWRF.

Table 3-2: Possible CV Openings of Bypass and Dump Line during Non-Operating of GWRF

Scenarios	GW HW Flow setpoint, m ³ /h	Bypass FCV (CV %)	Dump line FCV (CV %)
4) GW Non-operating hours	0	FCV1: 0% FCV2: 0%	90-100% (105 m ³ /h)

P6A.3.12 HEX Start-up

The heat exchanger temperature control valves shall be in AUTO, as a permissive for THP unit to start-up. There should be an option to temporarily bypass or override the first stage cooler's temperature controller.

It is expected that the THP unit start-up sequence initiates a nominal cooling water flow through the heat exchanges before hydrolysed sludge flow starts. The temperature control valve shall seamlessly starts modulating with controller in auto once the hydrolyse sludge starts flowing through.

P6A.3.13 HEX Shutdown

It is expected that the THP unit shut-down sequence shall allow the hydrolysed sludge flow to fully stop before closing the heat exchanger temperature control valves.

The first stage heat exchanger's bypass flow controller shall simultaneously take over the control and maintain the hot water supply to GW while the steam sparger shall be initiated to maintain the temperature control (as described in P6A.3.10).

P6A.3.14 HS Digester feed pumps and recirculation pumps

N.B. The following control description is indicative of the level of automation required. The Contractor shall provide a control philosophy from the specialist supplier.

The HS digester feed pumps are linked to the THP with two (2) pumps per THP unit. The HS digester feed pumps will be assigned as DUTY1, DUTY2, STANDBY1 or STANDBY2, i.e., corresponding to the status the associated THP units. If one DUTY pump fails, the related THP unit HS feed flow setpoint shall be trimmed automatically to the turndown flowrate. The Operator shall be alerted immediately and shall make the decision to switch over to the STANDBY THP unit.

The HS digester feed pumps shall be speed controlled based on the individual THP unit's HS feed flow setpoint. If HS cooler pumps are introduced in series with HS digester feed pumps, the controls shall be properly integrated to ensure adequate NPSH.

Ideally, the total HS feed flow setpoint shall be equally divided to the no of THP units IN SERVICE, however it is expected that the individual THP real time HS feed flow setpoint could be trimmed by the individual THP output vessel's level, as illustrated below and as described in P6A.5.4b) Trim on Upstream Level.

Table 3-3: Total HS Feed Flow Setpoint with respect to Number of THP Units in Service

Number of THP units IN SERVICE	Total HS Digester feed flow setpoint m³/h	THP unit HS feed flow setpoint (calculated) m³/h	Example of THP unit HS feed flow setpoint (actual), m³/h
1	16	16	17
2	32	16	16, 16
3	63	21	21, 22, 20
4	64	16	16, 15, 16, 17

The total HS Digester feed flow setpoint shall be trimmed by the upstream DWAS silo level as well, however the individual THP unit's vessel levels and capacity shall take precedence. The control details shall be provided by THP specialist.

The total HS feed flowrate and setpoint shall be calculated and displayed on SCADA. Additionally, the calculated % dry solids of the Hydrolysed Sludge stream shall be available for monitoring as well.

The HS digester feed recirculation loop contains three (3) digester HS recirculation pumps located at the common discharge line of the HS digester feed pumps at the THP courtyard.

They are required to maintain the momentum of the sludge around the HS digester feed recirculation loop. This prevents hydrolysed sludge from settling in any single point of the loop where low velocities are anticipated. The required recirculation rate and recirculation control shall be confirmed by the THP specialist. Due to the recirculation configuration, the recirculation pumps also act as booster pump to overcome the head loss across the long-distance digester loop.

The configuration shall be as follows:

- With recirculation (*Total pump flowrate >100m³/h*): Duty 1/ Duty 2/ Standby 1.
- Without recirculation flowrate (*Total pump flowrate <70m³/h*): DUTY/ Standby/ Maintenance.

Due to the long recirculation line, a dedicated short recirculation line could be provided at the THP courtyard from the discharge side of the recirculation pumps to the suction header. This will assist the start-up the HS recirculation pumps which are in series to the HS digester feed pumps, especially during the initial start-up of the THP system.

The pumps will operate whenever the THP is in service. The HS recirculation pump shall be of VFD type and is anticipated to operate in either flow or differential pressure mode. The backpressure valve shall maintain a certain recirculation flowrate through the backpressure control. The control of the feed and recirculation line shall also consider the digester sequential feed system. Depending on the recirculation control mode, the recirculation pump or backpressure valve control setpoints may be required to alternate based on the specific Digester being fed during feed cycle. E.g. When the nearest digester is being fed with HS, the recirculation pump may operate at a lower discharge head, when the further digester is being fed, the recirculation pump may operate at a higher discharge head. These settings shall be adjustable in SCADA to optimize the plant performance.

It is expected that the HS recirculation pumps can be manually bypassed temporarily if required for maintenance, and the Hydrolysed Sludge feed shall not be affected. The Hydrolysed Sludge Feed pumps and one recirculation pump shall be able to deliver the hydrolysed sludge from THP courtyard to the digesters without recirculation flow. The maximum allowable duration for turndown recirculation flow shall be advised by the THP specialist. If required, fall back strategy such as temporary dilution shall be advised by the THP specialist and implemented in the design as provision.

P6A.3.15 THP Process Gas

Process gas discharged from the flash tanks of the THP process units shall be cooled to less than 50 °C and shall be moisture free before routed to the Digesters separately. The THP system shall not allow discharge of Process Gas when high temperature alarm detected. THP system is encouraged to be optimized by recycling the flash gas for pre-heating to maximize the efficiency of the overall THP system.

P6A.3.16 THP Failure Mode

The Failure modes shall be defined in THP system's PCN. When individual THP unit fails, the alarms and trip signals should be available on SCADA.

A Cause & Effect chart shall be provided for developing the safeguarding and interlock trips related to the THP units. In the event any THP unit performs shutdown due to internal trips and becomes unavailable, Operator shall be prompted to confirm if the STANDBY THP unit shall be started.

P6A.3.17 THP Start-up

The THP system shall be started by manual initiation with the start-up sequence determined by the specialist supplier. The permissive to start THP shall be developed by the THP system specialist, it should include permissive checks for alarms. Additionally, the start-up permissive should check for availability of interface and supporting systems. This should include:

- Steam (Interface signal)
- DWAS Silo level (Interface signal)
- HS Digester feed pumps
- HS Recirculation pumps
- Discharge route to digester available (Interface signal)
- HS Cooling Heat Exchangers
- G2D Plant water
- Instrument Air

The start-up of the overall THP system shall consider the proper and safe start-up of interfaces as well. Particularly, the introduction of steam flowrate during THP start-up or capacity changes shall be done in a safe manner.

P6A.3.18 Shutdown

The individual THP units shall be shut down by manual initiation with the shutdown sequence determined by the specialist supplier.

The THP system should shutdown automatically in Emergency Scenario defined by THP's PCN. As a minimal, this shall include:

- i) Emergency shutdown signal from TWRP overall plant system
- ii) Steam unavailable
- iii) Both Silos are empty
- iv) Discharge route unavailable

The shutdown sequence shall be developed for specific scenario and the shutdown shall be done in a safe manner while minimising the inventory in the THP vessels (where possible).

P6A.4 DIGESTER TEMPERATURE

P6A.4.1 Overview

The bulk temperature of each digester shall be monitored from the temperature transmitters located at the suction of the individual digester's mixing pumps. Typical temperature for Mesophilic digesters is in the range of 36-40°C.

Assuming that the digester mixing is operating normally (i.e., achieving homogenous mixing), the bulk digester temperature of each digester should not exhibit any instantaneous changes due to the sequential feeding as described in P6A.5. There should only be minor temperature variation due to the change in ambient temperature across day and night-time.

The bulk (or steady state) temperature of the digesters can be estimated from the heat and material balance based on the feed ratio and temperature of individual sludge feed stream. Therefore, any change in the feed ratio or temperature of the individual feed stream could affect the overall temperature of the digesters. Other factors which could affect the Digesters' temperature are changes in average diurnal ambient temperature due to weather.

TPS and FW are fed at the room or ambient temperature as there are no heating involved upstream. GW from holding tank would typically be at a temperature in between 35 to 40°C. The only high temperature sludge feed stream which could contribute to the digester temperature is thermally hydrolysed sludge (HS).

HS (Hydrolysed Sludge) plays an important to maintain the digester' steady state/bulk temperature at the desired mesophilic temperature. Since the Hydrolysed Sludge from the THP process exits at elevated temperature of 110°C, it must be cooled by the Hydrolysed Sludge Coolers to a desired temperature before being fed into the digesters (approximately 50 -60°C).

The Hydrolysed Sludge cooled temperature setpoint shall be cascaded from the Digesters bulk temperature controller. Care shall be taken as part of the programming and PID tuning to ensure that the setpoint should not change frequently or drastically. However, the controls shall allow gradual setpoint trim based on periodic feedback on the overall bulk digester temperature controller.

P6A.4.2 Digesters' Bulk temperature

The individual digesters' bulk temperature shall be calculated based on the average of the two (2) temperature transmitters. For example, the bulk temperature of Digester no.1 would be:

$$T_{\text{digester1}} = 0.5 * (T_{113\text{-TIT-0101}} + T_{113\text{-TIT-0103}})$$

Alarm shall be provided if both readings deviate from one another. In the occasion that one (1) out of two(2) sensors is required to be taken offline for calibration or service, the Operator should be able to select only one of the temperature sensors as the input for the bulk temperature calculations. Deviation in the two readings could also indicate possibility of inefficient mixing. Additionally, the individual digester's bulk temperature should be similar, especially when operating in the default feeding cycle (GW and FW feed to all digesters). Operator should also be notified if any anomalies in individual digester is detected, as this will affect the overall digester system bulk temperature calculation.

The overall digesters' bulk temperature shall be calculated as follows:

$$T_{\text{bulk_digester s}} = (T_{\text{digester1}} + T_{\text{digester2}} + T_{\text{digester3}} + \dots) / (\text{No of Digesters online})$$

The time-averaged readings for each digester shall be easily monitored on the SCADA trends, e.g., hourly, 6-hourly, 24-hourly average etc.

P6A.4.3 Digesters' Bulk temperature controller

The bulk temperature setpoint of the Mesophilic Digesters ($T_{\text{bulk_digester_setpoint}}$) shall be set manually in SCADA. The optimum digester temperature shall be determined during commissioning and may be adjusted when there is a change in feed.

The setpoint will be used by the Digester's bulk temperature controller to provide a feedback to trim the Hydrolysed Sludge cooler temperature setpoint ($T_{\text{HS_setpoint}}$). The setup of the controller settings shall be done with care to ensure that any changes shall be incremental and shall not result in sudden shock in the digester feed.

When the digesters' bulk temperature controller is in auto, it will adjust the setpoint of the Hydrolysed Sludge outlet temperature until the bulk temperature of the digesters is equal to the desired setpoint.

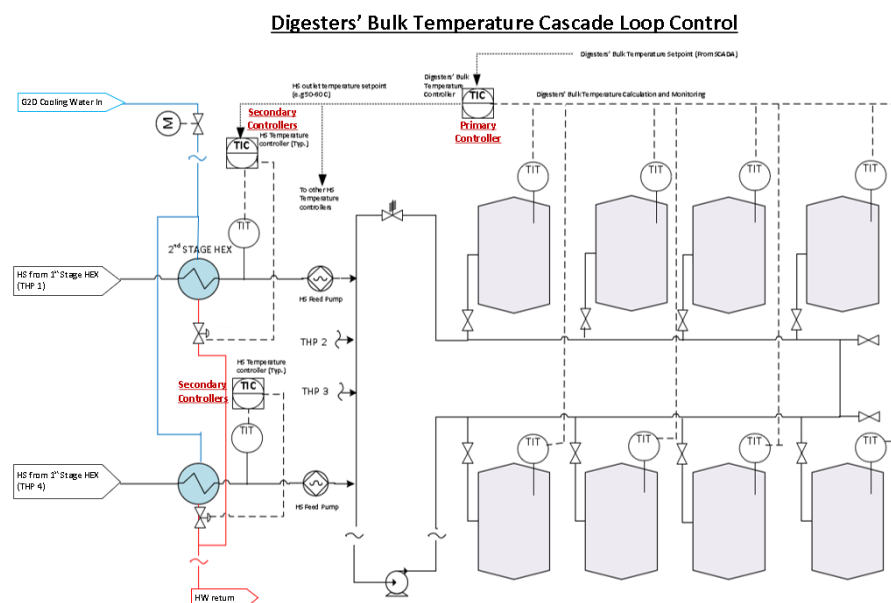


Figure 4-1 Proposed schematic for Digesters' Bulk Temperature Cascade Control (Typical for each THP train) under default feeding mode

Due to volume of each Digester and the long residence time, it is recommended that the PID controller shall react slowly. Any changes to the hydrolysed sludge cooler setpoint shall be based on analysing of digester bulk temperature trends between few hours to few days. The rate of change in digester's bulk temperature shall not exceed 1°C per day. It may be required to input historical data correlation based on other factors affecting the digesters' temperature such as variation due to ambient temperature. I.e., the controller shall be programmed to analyse and recognise disturbance due to other factors.

The time averaged deviation between the digesters' bulk temperature and the setpoints shall be the measured disturbance cascaded for trimming heat exchanger outlet temperature setpoint. As shown in Figure 4-1, the Digesters' Bulk Temperature controller shall be the Primary control loop while the HS second stage coolers outlet temperature controllers form the Secondary control loop. When Digesters' bulk temperature controller is in Auto, it calculates the Hydrolysed Sludge outlet temperature setpoint which shall be input for HS second stage coolers' outlet temperature controller. Option shall be provided on SCADA to operate the digester temperature control in manual mode, this allows the Hydrolysed Sludge temperature setpoint to be set by Operator during commissioning or troubleshooting time. As the relationship between heat input and the digester's temperature is non-linear, a good understanding of the digester's process is required to monitor the temperature control and tuning of the PID settings may be required from time to time.

As described in P6A.5 there are two (2) modes of digester feeding, the default one being all digesters receive same feed flow and ratio, the second one being FW and GW being fed to specific digesters only. Since the heat and material balance are different for FW and GW digesters, it would be required to control the Hydrolysed Sludge temperature for these two (2) types of digester separately. Additional logic can be implemented to allow the $T_{HS_setpoint}$ to be alternating based on the feeding cycle described in P6A.5. I.e., the $T_{HS_setpoint}$ would be expected to automatically switchover from one setpoint to another when the HS is step feeding from a GW to a FW digester.

To minimise the number of setpoint changes, the same type of digesters shall be fed in sequence (further described in P6A.5.) Figure 4-2 provides the example proposed controls during the different digester feeding mode (mode 2).

Digesters' Bulk Temperature Cascade Loop Control (Digester Feeding Mode 2)

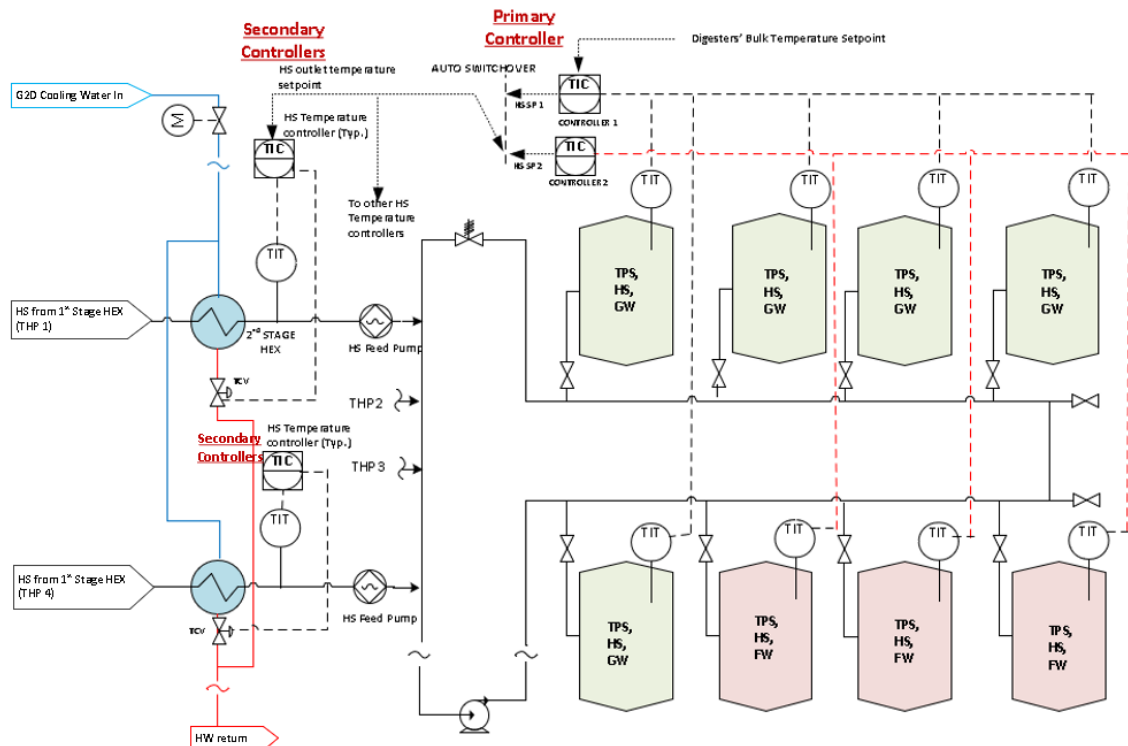


Figure 4-2 Proposed schematic for Digesters' Bulk Temperature Cascade Control (Typical for each THP train) under digester feeding mode #2

Table 4-1 Example of digester feed stock ratio and temperatures; and the explanation on heat & mass balance calculation based on the inputs below to obtain the $T_{HS_setpoint}$.

Design-MML with PSD to THP						
	Mode 1: (default)		Mode 2:			
No in Service	10 Digesters		4 FW digesters		6 GW Digesters	
	Feed Flow Ratio	Feed Temp(°C)	Feed Flow Ratio	Feed Temp (°C)	Feed Flow Ratio	Feed Temp(°C)
TPS	51.4%	30.0	45.7%	30.0	56%	30.0
HS	30.1%	52.1 ($T_{HS_setpoint}$)	26.8%	57 ($T_{HS_setpoint_1}$)	33%	49 ($T_{HS_setpoint_2}$)
GW	6.2%	40.0	0.0%	40.0	11%	40.0
FW	12.4%	30.0	27.5%	30.0	0%	30.0
Calculated Digester's Feed Temperature (Average)	37.3 °C		37.4°C		37.4°C	

Digester's Bulk Temperature ($T_{\text{bulk_digester_setpoint}}$)	36.0 °C	36.0°C	36.0°C
<u>Estimation of $T_{\text{HS_setpoint}}$</u> The $T_{\text{HS_setpoint}}$ can be calculated based on heat and mass balance with assumptions on the heat loss to ambient temperature (and negligible heat gain from solar radiation and exothermic reactions of digestion). The above example assumes 30kW of heat loss through the tank walls. The feed temperature from TPS and FW are assumed to be ambient temperature. The temperature of GW (envisaged to be 35-40°C) can be obtained from the temperature sensor at the GW holding tank. The heat balance across the digester is estimated using the following equation: $Q_{\text{dig.heat_in}} + Q_{\text{dig.heat_out}} + Q_{\text{dig.heat_loss(tank+piping)}} + Q_{\text{exothermic}} = 0$ Where, $Q_{\text{dig.heat_in}} = \sum (M_{\text{TPS}} \times T_{\text{TPS}} \times C_{\text{TPS}} + M_{\text{HS}} \times T_{\text{HS}} \times C_{\text{HS}} \dots)$, $Q_{\text{dig.heat_out}} = M_{\text{DS(DigestedSludge)}} \times T_{\text{DS}} \times C_{\text{DS}}$ and $Q_{\text{dig.heat_loss(tank+piping)}}$ is estimated to be 30kW and $Q_{\text{exothermic}}$ is assumed as negligible. The mass flow M shall be calculated based on sludge flowmeter readings and sludge density input from laboratory results. The specific heat capacities of various feed sludge are assumed to be the same. In steady state condition, the digested sludge temperature is assumed to be the same as Digester's bulk temperature. The equation could then be rearranged to obtain the temperature setpoint for Hydrolysed Sludge $T_{\text{HS_setpoint}}$. The digester controller shall be capable of calculating the HS temperature based on the feed ratio and mass balance, the value shall be displayed on SCADA as well. Individual trim factor shall be applied to each term in the equation, this shall be the basis for the proposed feedback cascade controller. The trim factors weighting factor to each term may be adjustable. When the controller is in auto mode, the factors will be manipulated based on the disturbance measured for digester bulk temperature trend. If the live measurement of the surrounding air temperature is available to the PLC, this could be used as an optional input to the sludge temperature and heat loss calculation for better prediction. On the other hand, when Digester Bulk Temperature controller is set to Manual, the estimated Digester's bulk temperature (from T_{DS}) will be calculated and displayed to assist the Operator in determining the desired HS temperature setpoint.			

The allowable limit of the $T_{\text{HS_setpoint}}$ is envisaged to be in the range of 50 to 60 °C. The high limit of the of the setpoint based on the mass balance is about 60°C (assuming digester mixing ratio of 1:3), to avoid high instantaneous temperature at the respective digester mixing pump discharge line (feed nozzle). This limit setpoint shall be confirmed based on the design and operational digester mixing pump flow to HS flow ratio configured during commissioning. The aim is to keep the comingled stream (HS plus recycled digested sludge stream) temperature not more than 10°C higher than the bulk temperature of the digester, as illustrated in Figure 4-3 below. This can be monitored from the temperature transmitters located at the discharge line of the digester mixing pump (e.g., for Digester no.1, the tag numbers are 113-TIT-0102 and 113-TIT-0104). In case high temperature is detected during feeding cycle, the HS feed pumps shall be interlocked to reduce flow setpoint or stop completely with high-high trip.

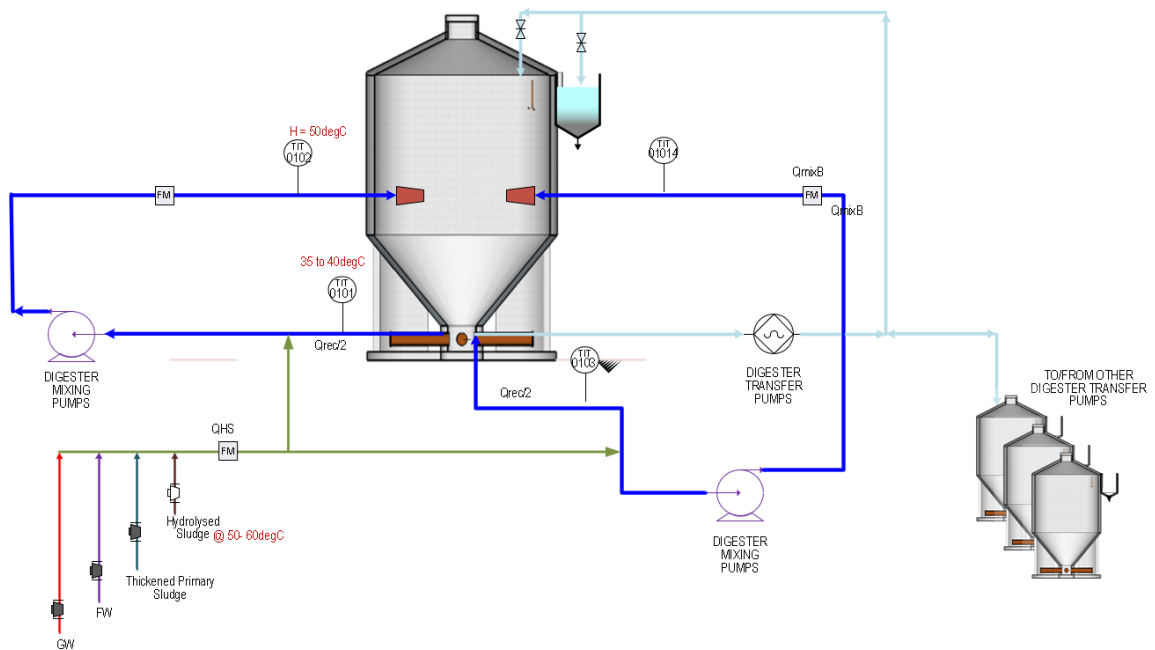


Figure 4-3 Individual Digester Feed Schematics (Example of temperatures during HS feeding sequence)

P6A.5 DIGESTER FEED AND DISCHARGE

The following is an excerpt from Biosolid Plant Digester's PCN (also provided in Volume 6). It describes the Digesters feeding arrangement and sequence control philosophy.

P6A.5.1 Overview

The mesophilic anaerobic digesters are fed with the following feed streams:

- Thermally Hydrolysed Sludge (HS), (pumping by others).
- Thickened Primary Sludge (TPS).
- Food Waste (FW);
- Greasy Waste (GW).

The feed streams are co-digested to generate biogas and to produce energy. The biogas generated is transferred to the IWMF. Biogas buffering capacity is provided on-site via the biogas holders.

The anaerobic digesters at TWRP are silo shaped concrete tanks with fixed roof. The digesters operate in a fill and spill mode and maintain a constant digestate sludge level adjusted by the telescopic bell mouth valve.

Thermally hydrolysed sludge is cooled using hydrolysed sludge cooling heat exchangers (by others). The cooled hydrolysed sludge is then transferred to the hydrolysed sludge feed ring main to the digesters. Thickened primary sludge, food waste and greasy waste is pumped to the respective sludge feed ring main from the thickened primary sludge holding tanks, food waste holding tanks and greasy waste holding tanks respectively. Thickened primary sludge holding tanks, food waste holding tanks and greasy waste holding tanks are provided to attenuate flow and load to the digesters and to provide buffer capacity. DWAS cake silo is provided prior to the thermal hydrolysis process to attenuate the flow and load to the thermal hydrolysis process and subsequently to the digesters.

Each digester is provided with pump mixing system with sludge recirculation loop. The hydrolysed sludge, thickened primary sludge, food waste and greasy waste are mixed with the digestate at the suction of the sludge mixing pump to provide instant mixing of feed sludge with digestate content and to avoid heat shocks.

Hydrolysed sludge and thickened primary sludge shall be distributed equally among IN SERVICE mesophilic anaerobic digesters; however, the operator shall be able to choose number of digesters that can receive food waste and greasy waste or both.

Distribution of each feed stream to the digester shall be controlled by actuated eccentric plug valves. One dedicated eccentric plug valve is provided for each digester, per feed stream (4 eccentric plug valves per digester) to control the feed duration.

The digested sludge overflows through two actuated telescopic valves, one drawing from the bottom of the digester and one from the top, to the DS holding tanks ring main. Normally, one telescopic valves will be in a set position for overflowing, the other will be fully raised.

Digesters 5 and 10 have are not included within the scope of this Contract. Space and connection points shall be provided to allow these digesters, and the associated MEICA plant, to be installed in future. These digesters shall be set as OUT OF SERVICE on SCADA.

P6A.5.2 Control Objectives

TWRP digester feed shall be fully automated with minimum operator intervention. The control system design shall achieve the following:

- Achieve consistent feed to each individual anaerobic digester and uniform distribution of feed solids with a flexibility to reduce the feed to a particular digester if the VFA / Alkalinity ratio increases beyond a trigger point;
- Provide flexibility in control system to enable operator to select the number of digesters receiving FW and GW;
- Allow level in the feed tanks (TPS feed tank, DWAS cake silo, FW feed tank, GW feed tank) to vary to attenuate flow and load to the digesters;
- Reduce manpower through higher-level plant automation;

P6A.5.3 Operator Settings

All digester feed pumps are assigned as DUTY/, STANDBY/ MAINTENANCE. The duty assignments of each pump will be automatically assigned in line with the overall duty rotation strategy. There will also be the facility to manually select the duty assignments from the SCADA.

The operator shall set:

- The individual digesters IN SERVICE;
- The digester feed % flow distribution between each operating digester for TPS, HS, FW, and GW (buttons are also provided to reset each feed to the default even distribution) – Note the flow is varied when pumping to a digester.
- Digester feed cycle duration;
- The digester discharge rotation time;
- The operating level of the digesters.

P6A.5.4 Normal Operation

P6A5.4.1. Digester Feed Pump Control

Each of the feed stream flows to the digesters are controlled using the feed pump speed and number of feed pumps operating for that stream, these are adjusted to maintain a feed flow setpoint using feedback from the associated feed pump discharge flow meter.

Based on the measurement of the digester feed stream flow compared to the setpoint for each feed stream, the number of feed pumps operating and VFD speed shall be controlled. The feed pump speed shall be regulated between the minimum and maximum speed. These speed limits shall be hardcoded for the physical limits of the pump and take precedence over all other control functions and commands.

If there is more than one feed pump operating, on a particular feed stream, all pumps should be operating at the same speed.

If measured digester feed flow for each feed stream is less than “digester feed flow setpoint” minus dead band for respective feed stream, start duty digester feed pump at minimum speed. If the duty feed pump is in operation, control loop shall adjust the speed of the duty digester feed pump to meet the “digester feed flow setpoint” for respective feed stream. If duty pump reaches to its maximum speed for greater than “duty feed pump operating at 100% speed delay time”, then start assist feed pump.

The duty and assist feed pump shall start operating at the “duty assist feed pump start speed”. The control loop shall adjust the speed of both pumps to meet the “digester feed flow setpoint” for respective feed stream.

If speed of two operating pumps is less than “duty, duty assist pump minimum speed” for greater than “duty, duty assist pump minimum speed delay time”, then stop the duty assist pump.

The duty pump shall start operating at 100% speed. The control loop shall adjust the speed of duty pump to meet the “digester feed flow setpoint” for a respective feed stream.

If speed of duty pump is less than “duty pump minimum speed” for greater than ‘duty pump minimum speed delay time”, then stop the duty pump.

P6A5.4.2. Digester Feed Flow Overall

a) Setpoints

The initial overall flow setpoint for each feed stream will be estimated as an average of past 7 days of flow for each individual stream to the digesters. The operator shall be able to alter the setpoint for each feed stream manually (m³/d).

b) Trim on Upstream Level

A trim of the overall feed stream flow setpoint will be able to automatically vary the overall feed stream flow setpoint based on the on level in the respective feed tanks – TPS, FW, GW and load measured at the DWAS cake silo. The tank/silo contents will be maintained within a range to utilise the tank/silo volume for attenuation.

Feed Stream Flow Setpoint Control for TPS, FW and GW

Normal operating level, between L1 and L2 setpoints (% full), for each feed tank (TPS, FW and GW) shall be entered into SCADA. The level in each tank shall be measured and compared against the normal operating level setpoint value for that tank. If actual measured level in the tank is within the normal operating minimum and maximum level setpoint (L1 and L2), then the digester feed flow setpoint for respective feed stream shall remain unchanged.

If measured level in the feed tank is above the normal operating level setpoint L2 but below L3 for an operator adjustable period, then the PLC shall adjust the digester feed flow stream setpoint proportionally. For example, if actual measured level in the TPS tank is above L2 for ten (10) minutes (operator adjustable), then increase the pump rate by:

$$\% \text{Increase in pump flow setpoint} = (L_{PV} - L_2) / (L_3 - L_2)$$

$$Q_{TPS} = Q_{TPSO} \times (1 + \% \text{increase in pump flow setpoint})$$

Where:

L_{PV} – process value of level in the tank in % full

Q_{TPSO} – Original flow setpoint for TPS

Q_{TPS} – New flow setpoint for TPS

L0, L1, L2, L3 – are level setpoint in % full.

If actual measured level in the TPS tank is below L1, but above L0 for ten (10) minutes (operator adjustable), then decrease the pump rate by:

$$\% \text{Reduction in pump flow setpoint} = (L_{PV} - L_1) / (L_1 - L_0)$$

$$Q_{TPS} = Q_{TPSO} \times (1 - \% \text{Reduction in pump flow setpoint})$$

The normal operating minimum and maximum level setpoint shall have a wide range to provide stability in operation.

If level in the tank drops below L0 for 10 minutes (operator adjustable), then stop the respective digester feed pump.

If level in the tank rises above L3 for 10 minutes (operator adjustable), then operate the respective digester feed pump at maximum flow, which is an operator adjustable flow setpoint.

Overall feed flow for each of the fluid streams is adjusted depending on the level in the source tank.

This is illustrated in the table below:

Table 5-1: Digester Feed Pump Actions

<i>Level</i>	<i>Actions</i>	<i>Flow Regime</i>
L3	Critical Alarm – Digester Feed Stream Flow at Max	FR3
L2	Digester Feed Flow Stream Setpoint increased proportional to level above L2 (between normal flow setpoint and maximum pump flow)	FR2
	Digester Feed Stream Flow Setpoint Remains Unchanged	FR1
L1	Digester Feed Stream Flow Setpoint decreased proportional to level below L1 (between normal flow setpoint and minimum pump flow)	FR0
L0	STOP Digester Feed Pumps	

Feed Stream Flow Setpoint Control for HS

It is envisaged the flow setpoint for the HS will be trimmed in the same manner using the weight recorded in the DWAS silos. However, the level in the vessel immediately upstream of the HS Digester Feed Pumps will also need to be used to adjust and stop the pumps on low level and may be used to trim the flow setpoint. The control of the Digester HS Feed Pumps shall be developed by the Contractor and THP supplier.

P6A5.4.3. Digester Feed Flow Individual

c) Flow Stream Selection

The default is that all feed flow streams are fed to all IN SERVICE digesters.

The operator can adjust this and select which digesters are to be fed with which flow stream. A 'default' button shall be provided in SCADA to reset to feed all flow streams to all digesters.

Note: the distribution can also be adjusted for the individual flow streams and digesters, see section f).

d) Duration

The Digester Feed Cycle Time will be set by operator input to the SCADA system.

HS, TPS, FW and GW feed duration to each digester shall be calculated based on selected digester feed cycle time (operator adjustable) and number of digesters in operation for respective feed stream.

$$\text{Digester Feed Cycle Duration for TPS/HS} = \frac{\text{Digester feed cycle time (say 120 mins)}}{\text{Number of digesters receiving TPS/HS}}$$

The digester feed cycle duration for TPS and HS will be same, as number of digesters receiving both HS and THP shall be same, this will normally be all the digesters IN SERVICE. If digester feed cycle time is 120 minutes, then the digester feed cycle duration for TPS and HS with 8 digesters available, will be 15 minutes each.

The default will be for the FW and GW to also be fed to all digesters IN SERVICE and the digester feed cycle duration for FW and GW will be calculated as above. This arrangement is shown schematically below:

Example of Default mode: All digesters in service, 120 min feed cycle time, FW/GW to all digesters

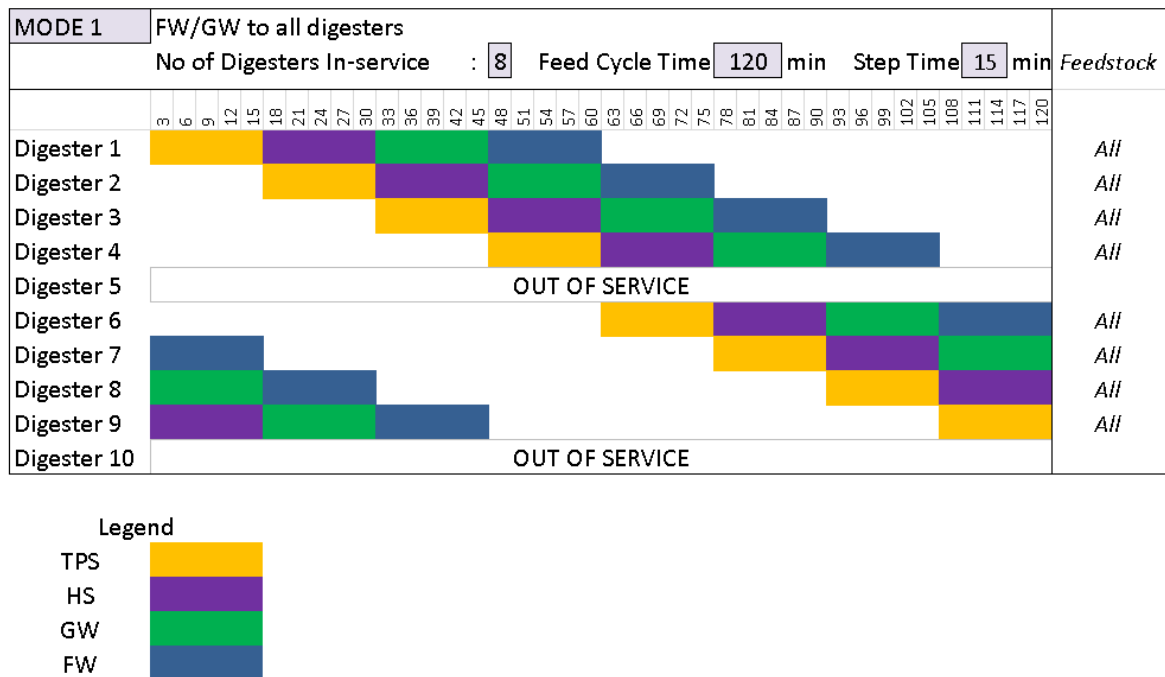


Figure 5-1 Schematics of Digester Feed Cycle in Default Mode

There may be times when the number of digesters receiving either GW or FW will be required to be reduced. For example, if 5 digesters are selected to receive GW and 3 selected to receive FW, then the digester feed durations for GW and FW will be 24 minutes and 40 minutes respectively. This arrangement is shown schematically below:

Example: All digesters in service, 120 min feed cycle time, FW to 3 digesters, GW to 5 digesters

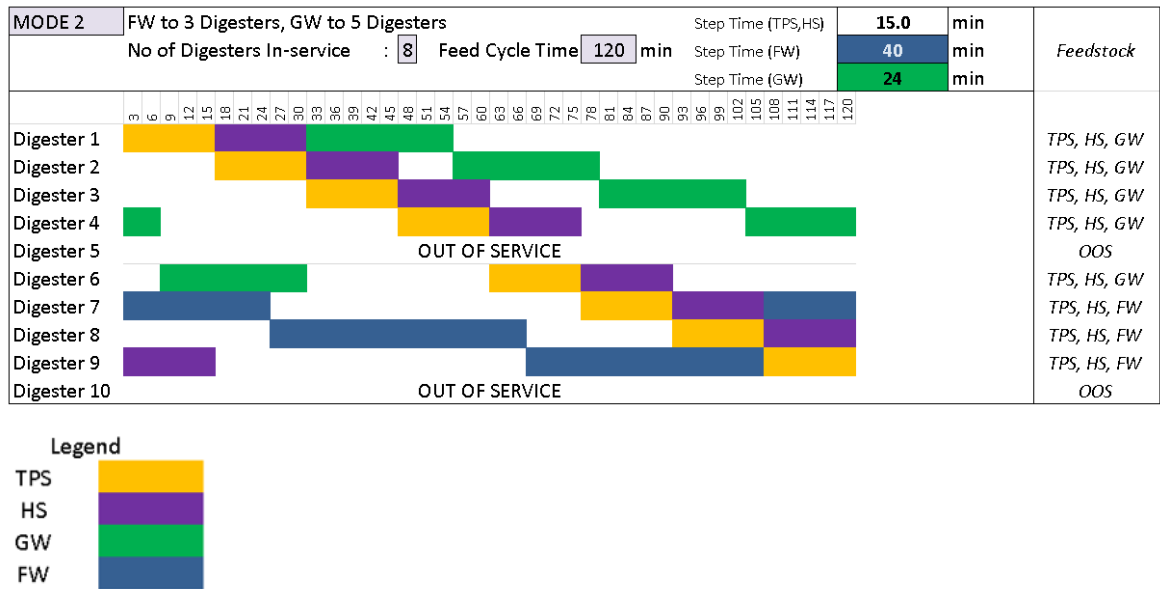


Figure 5-2 Schematics of Digester Feed Cycle in when All Digesters In Service

As shown in the example above, Digester 1,2,3,4,6,7 which receive GW are fed in sequence followed by Digester 7,8,9 which receive FW, this allows the HS temperature setpoint to be changed only twice in a cycle as described in P6A.4.3. A check shall be included when setting the Digester Feed Cycle Time and the number of digesters receiving the feed streams to ensure that the sum of the feed stream durations does not exceed the Digester Feed Cycle Time.

e) Feed Valve Operation

The digesters shall be fed in sequence, normally starting with Digester 1.

Digester 1 shall start with a TPS feed duration by opening the associated eccentric plug TPS feed valve for the calculated duration. At the end of the Digester 1 TPS feed cycle duration, the TPS feed valve for the Digester 2 will open, and be confirmed open, before closing the TPS feed valve for Digester 1.

Followed by TPS feed duration, Digester 1 shall initiate HS feed duration, followed by FW feed duration (if applicable) and then GW feed duration (if applicable).

At any given time, each feed stream (TPS, HS, FW, GW) shall feed to only one digester; and one digester shall receive only one feed stream at any given time.

Should a valve fail to open fully within a set time period, the feed valve on the preceding tank will stay open for the duration of that tanks feed cycle. Once the feed valve on the subsequent tank is confirmed open, both feed valves shall close.

Should a valve fail to close fully within a set time period the feed cycle should continue on the assumption that it is closed. However, an alarm will be raised for the valve.

The status of the feed valve on each digester shall be used to measure the volumetric flow of each fluid to each digester.

The feed volume to each digester shall be measured and displayed as the current value as well as: a rolling 24-hour average, i.e., the volume fed to the digester in the last 24 hours; the 24-hour and 20 day (adjustable) rolling average of the hydraulic retention time.

If level in any of the feed tanks drops below “low level” setpoint, raise an alarm and stop the respective feed pump. The feed duration timer continues, and the valve closes that the end of the duration. Subsequent feed valves for this stream shall not open but the feed duration shall be allowed to expire before the next feed stream is opened.

If the flow in a digester mixing/recirculation loop is not detected, then the feed streams to that digester must stop.

f) Trim on Individual Loading Requirements

The default is that all digesters receive the same flow. To enable the operators to adjust the loading of individual digesters, due to poor digester health or the requirement to FW and GW to particular digesters, the feed flow setpoint for a feed flow stream is adjusted for individual digesters while that digester is being fed with that particular feed.

The operator shall be able to adjust the distribution of feed to each digester by trimming the feed flow setpoint (a)). Below is an example of the default equal distribution of TPS, HS, FW and GW if 8 digesters are available, and each digester is receiving FW and GW. The predicted resulting HRT, organic VS loading and the specific energy loading shall be calculated and displayed.

Table 5-2: Example of Default Equal Distribution of TPS, HS, FW and GW

<i>Digesters</i>	<i>TPS feed [%]</i>	<i>HS feed [%]</i>	<i>FW feed [%]</i>	<i>GW feed [%]</i>	<i>HRT[†] [d]</i>	<i>Organic Loading* [kgVS/m³.d]</i>	<i>Specific Energy Loading* [kgCOD/d.kgVS in digester]</i>
Digester 1	1.00	1.00	1.00	1.00			
Digester 2	1.00	1.00	1.00	1.00			
Digester 3	1.00	1.00	1.00	1.00			

<i>Digesters</i>	<i>TPS feed [%]</i>	<i>HS feed [%]</i>	<i>FW feed [%]</i>	<i>GW feed [%]</i>	<i>HRT† [d]</i>	<i>Organic Loading* [kgVS/m³.d]</i>	<i>Specific Energy Loading* [kgCOD/d.kgVS in digester]</i>
Digester 4	1.00	1.00	1.00	1.00			
Digester 5	0.00 O	0.00	0.00	0.00			
Digester 6	1.00	1.00	1.00	1.00			
Digester 7	1.00	1.00	1.00	1.00			
Digester 8	1.00	1.00	1.00	1.00			
Digester 9	1.00	1.00	1.00	1.00			
Digester 10	0.00 O	0.00	0.00	0.00			
Total ‡	8.00	8.00	8.00	8.00			

†Calculated

*Calculated from values retrieved from the laboratory management system. The number of results used TBC

‡ This is the number of digesters selected for that feed flow stream

M Manual override of even load distribution

O Out-of-Service taken from the digester status – all inputs are disabled

Below is an example of distribution of TPS, HS, FW and GW if 8 digesters are available and the operator has decided to manually override the even distribution and reduce the load (flow) to digesters 2 and 6. The operator shall enter these manual values and a button shall be provided to set the proportions to evenly divide the remaining flow between the remaining IN SERVICE digesters and selected for that particular flow stream, that are not overridden. The predicted resulting HRT, organic VS loading and the specific energy loading shall be recalculated and displayed.

Table 5-3: Example of Manual Override Distribution of TPS, HS, FW and GW (Reduced Flow to Digesters 2 and 6)

<i>Digesters</i>	<i>TPS feed [%]</i>	<i>HS feed [%]</i>	<i>FW feed [%]</i>	<i>GW feed [%]</i>	<i>HRT† [d]</i>	<i>Organic Loading* [kgVS/m³.d]</i>	<i>Specific Energy Loading* [kgCOD/d.kgVS in digester]</i>
Digester 1	0.80 M	0.80 M	0.80 M	0.80 M			
Digester 2	1.07	1.07	1.07	1.07			
Digester 3	1.07	1.07	1.07	1.07			
Digester 4	0.80 M	0.80 M	0.80 M	0.80 M			
Digester 5	0.00 O	0.00 O	0.00 O	0.00 O			
Digester 6	1.07	1.07	1.07	1.07			
Digester 7	1.07	1.07	1.07	1.07			
Digester 8	1.07	1.07	1.07	1.07			
Digester 9	1.07	1.07	1.07	1.07			
Digester 10	0.00 O	0.00 O	0.00 O	0.00 O			
Total ‡	8.00	8.00	8.00	8.00			

†Calculated

*Calculated from values retrieved from the laboratory management system. The number of results used TBC

‡ This is the number of digesters selected for that feed flow stream

M Manual override of even load distribution

O Out-of-Service taken from the digester status – all inputs are disabled

Below is an example of distribution of TPS, HS, FW and GW if 8 digesters are available and 5 are set to receive GW and 3 to receive FW.

Table 5-4: Example of Distribution of TPS, HS, FW and GW (Digester 5 to receive GW, Digester to receive FW)

<i>Digesters</i>	<i>TPS feed [%]</i>	<i>HS feed [%]</i>	<i>FW feed [%]</i>	<i>GW feed [%]</i>	<i>HRT† [d]</i>	<i>Organic Loading* [kgVS/m³.d]</i>	<i>Specific Energy Loading* [kgCOD/d.kgVS in digester]</i>
Digester 1	1.00	1.00	1.00	0.00			
Digester 2	1.00	1.00	1.00	0.00			
Digester 3	1.00	1.00	0.00	1.00			
Digester 4	1.00	1.00	0.00	1.00			
Digester 5	0.00 O	0.00 O	0.00 O	0.00 O			
Digester 6	1.00	1.00	1.00	0.00			
Digester 7	1.00	1.00	0.00	1.00			
Digester 8	1.00	1.00	0.00	1.00			
Digester 9	1.00	1.00	0.00	1.00			
Digester 10	0.00 O	0.00 O	0.00 O	0.00 O			
Total ‡	8.00	8.00	3.00	5.00			

†Calculated

*Calculated from values retrieved from the laboratory management system. The number of results used TBC

‡ This is the number of digesters selected for that feed flow stream

M Manual override of even load distribution

O Out-of-Service taken from the digester status – all inputs are disabled

■ Unavailable due to digester flow stream selection

P6A5.4.4. Digester Discharge

The speed of movement of the discharge telescopic valves may need to be limited to ensure a positive pressure is maintained in the digesters at all time.

Normally, the draw off from the digesters shall be from the bottom with the bottom draw-off valve being at the set operating position. When activated, the withdrawal rotation will automatically change the withdrawal to the top telescopic valve to allow the withdrawal of the top lighter fraction through the top withdrawal. After a set time duration (default 10 hours), the withdrawal will change back to the bottom draw off.

The withdrawal rotation will occur sequentially between the digesters. The default cycle time will be 12 hours so that each the withdrawal is rotated every four days for each digester.

When a withdrawal rotation is started the bottom draw-off valve will be raised to fully closed. When confirmed closed the top withdrawal valve will lower to the set operating position. After the top withdrawal duration has expired the reverse occurs with the top withdrawal valve being raised to fully closed before the bottom withdrawal valve is lowered to the set operating position. This order of operation is to allow a flush of the pipe on flow commencing.

The withdrawal rotation of the next IN SERVICE digester will start after the cycle time has expired.

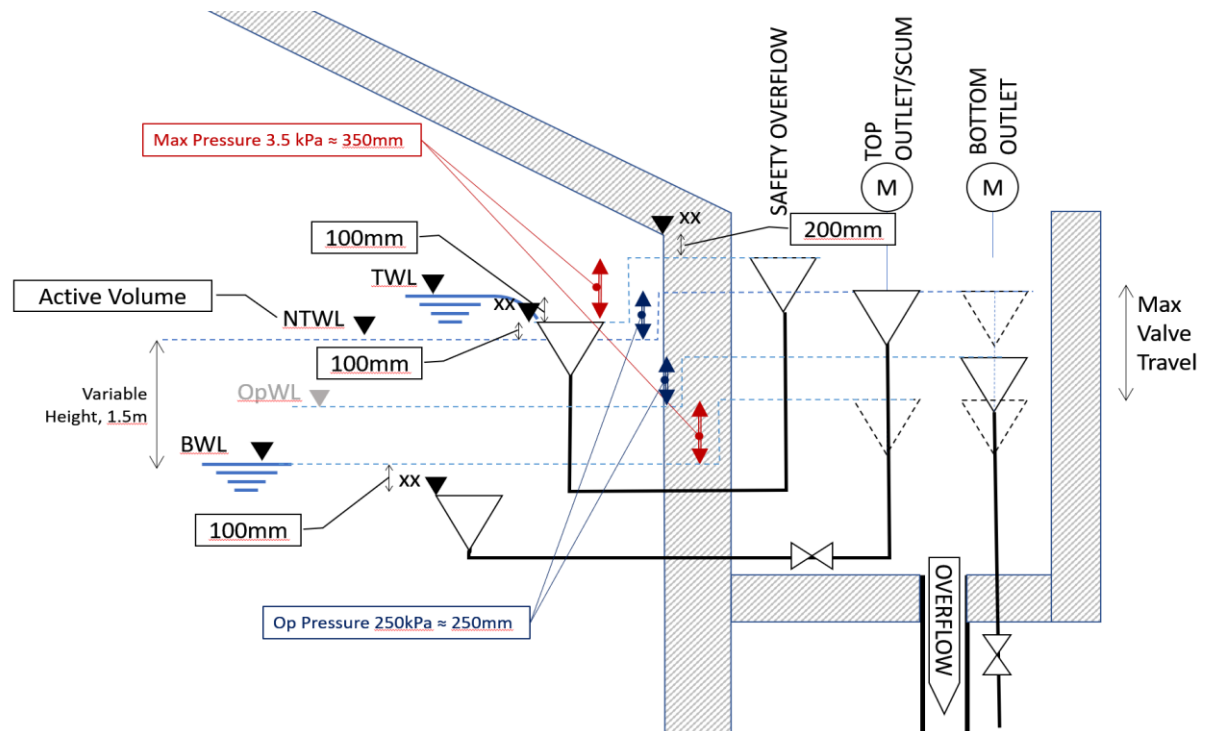


Figure 5-3: Diagram of the Level in the Digesters

P6A.5.5 Start-up

The start-up of the overall digester feed and discharge system shall be operator initiated.

The start-up of digester feed and discharge for each digester shall be operator initiated when a digester is put IN SERVICE. Feed pumps shall only be started if there is not a low level in the feed tank(s) and a discharge route is available, including a feed valve being open and the associated digester level not high.

P6A.5.6 Shutdown

The shutdown of the overall digester feed and discharge system shall be shut down by manual initiation.

The shutdown of digester feed and discharge for each digester shall be operator initiated when a digester is put OUT OF SERVICE.

Feed pumps shall be stopped if there is a low level in the feed tank(s) or a High-High in the receiving digester or a discharge route is not available. Also, if a High-High temperature is detected the receiving digester the HS digester feed pumps shall be stopped.

P6A.6 SIDESTREAM N-REMOVAL

P6A.6.1 Overview

The operation of the Sidestream N Removal Facility will be determined by the specialist technology supplier. The operation and control of the Facility must be fully integrated to achieve stable, efficient, and safe operation with minimal manual intervention.

The process control narratives and the functional design specifications shall be developed by the specialist technology supplier and be incorporated into the overall plant control system.

The purpose of the Sidestream N-Removal Facility is to treat centrate flows to minimise nitrogen loads returned to the liquid stream.

The Sidestream N Removal Facility is a vendor specific process that uses Anammox organisms to treat flows from the following sources:

- Digested Sludge Dewatering Centrate
- IWMF Dryer Condensate
- WAS Pre-dewatering Centrate

The system comprises the following sub-systems:

- Centrate flow diversions
- Flow equalisation, solids separation and deammonification reactor feed
- Deammonification reactors
- Selective sludge wasting
- Aeration system
- Discharge wet wells
- Sludge wet well

This section provides an overview of the control objectives for the Sidestream N Removal Facility.

P6A.6.2 Control Objectives

The Sidestream N Removal Facility shall be fully automated with minimum operator intervention. The control system design shall achieve the following:

- Achieve consistent feed to the deammonification reactors
- Solids separation to reduce solids loading on deammonification reactors
- Maximise nitrogen removal within the deammonification reactors
- Allow selective wasting to maintain the Anammox bacteria in the facility.
- Maintain safe, stable and efficient operation to achieve the degree of nitrogen removal required
- Reduce manpower through higher-level plant automation.

P6A.6.3 Operator Settings

The Sidestream N Removal Facility deammonification reactors are assigned as DUTY1, DUTY2, DUTY3, DUTY4. The assignment of each reactor will be manually selected from the SCADA.

P6A.6.4 Normal Operation**P6A5.4.5. Centrate Flow Diversion Control**

The centrate pumps shall be controlled externally to the facility..

The Sidestream N Removal Facility shall take all of the dewatering centrate flow, a bypass is provided for commissioning and emergencies.

The flow of thickening centrate shall be controlled to provide the required dilution flow to the Sidestream N Removal Facility with the remainder being transferred back to the TC pipeline going to the ILM.

These arrangements are illustrated schematically in Figure 6-1.

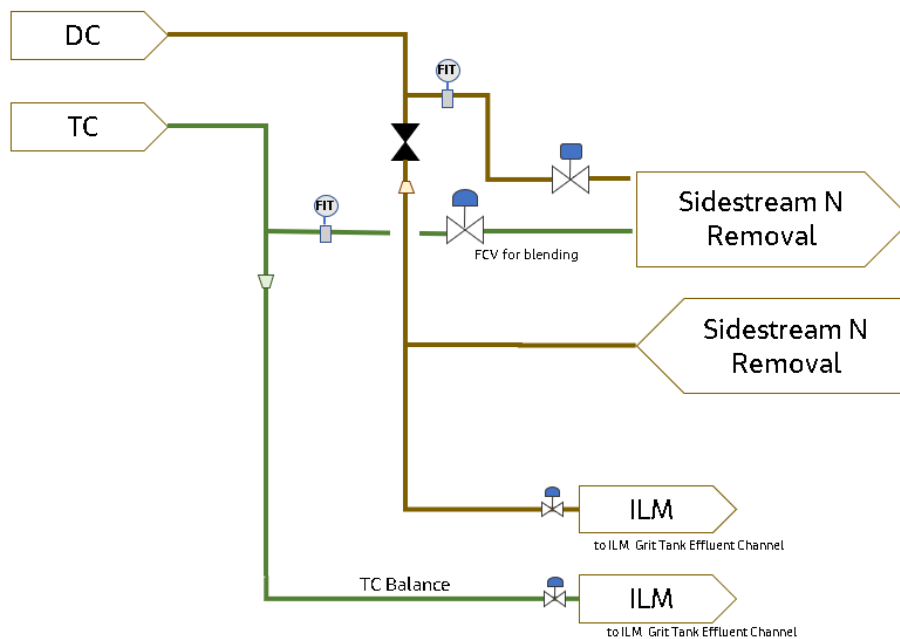


Figure 6-1: Schematic of the Flows in and out of the Sidestream N Removal Facility

P6A5.4.6. Flow Equalisation

Automatic control of the equalisation tanks is to be initiated by the control strategy employed by the specialist technology provider.

The equalisation tanks are equipped with an automatic flushing system to allow for periodic cleaning of accumulated solids from the tank floor. Initiation of the automatic flushing sequence will be based on programmed and operator adjustable parameters.

Discharge from the Sidestream EQ Tank will gravitate to the Deammonification Reactors. Flow distribution and feeding control from EQ tank to the individual reactors shall be determined by the specialist technology provider.

P6A5.4.7. Deammonification Reactors

Normal operation of the deammonification reactors will be in automatic mode with all reactors in service. The deammonification reactors will go through a cycle of feeding, aeration, mixing, sludge wasting, and discharge based on programmed and operator adjustable parameters.

P6A5.4.8. Sludge Wasting

Sludge Removal and recycling is an important step of the deammonification process and is controlled based on specialist technology supplier requirements. Heavy fraction Anammox sludge is to be separated and recycled to the deammonification reactors

P6A5.4.9. Aeration System

Automatic control of the blowers will be initiated by the control strategy employed by specialist technology provider.

The operator shall set individual blowers In-Service

P6A5.4.10. Discharge Wet wells

Treated centrate from the deammonification reactors will gravitate to the discharge wells at the end of each cycle. The sludge wasting shall also be diverted to the discharge wet well.

Flow from the discharge wells is pumped to the ILM grit tank effluent channel. Duty and Standby pumps are to be provided. The pumps are proposed to be operated based on the water level within the discharge wet well. For the pumps to be able to start, confirmation that a discharge route to the ILM grit tank effluent channel is available. Following the stop of a Duty pump, duty rotation should occur to equalise run times on the pumps.

P6A5.4.11. Sludge Wet Well

Provision is also made to direct anammox organisms to a sludge wet well for seeding to other plants. Anammox sludge is to be transferred via road tanker.

END OF SECTION